AGRISURE – Product Requirements Document (PRD)

1. Product Overview

AGRISURE is an AgTech platform designed to help smallholder farmers make better financial and production decisions before planting crops. The product combines crop yield simulation, input loan requests, and risk scoring into one simple decision-support tool. Farmers enter farm data such as crop type, farm size, and expected input investment, and the platform predicts potential harvest output and determines the financial risk of providing a loan.

2. Problem Statement

Smallholder farmers often lack structured financing and tools to estimate crop yield before investing in farming inputs. Financial institutions also struggle to evaluate agricultural lending risk due to limited reliable data. This results in underfunded farmers, unpredictable harvests, and higher loan default rates.

3. Target Users

Primary Users: Smallholder farmers managing farms between 1 and 10 hectares. Secondary Users (future): Agricultural lenders, cooperatives, government agriculture programs, and crop insurance providers.

4. MVP Product Scope

The MVP version focuses on a simple simulation experience allowing farmers to forecast yield and understand loan eligibility. The goal is to validate the core concept: that predictive insights can help farmers make better financing decisions and help lenders assess risk.

Feature	Description
Farmer Registration	Basic onboarding collecting farmer name, phone, location, and farm size.
Crop Selection	Farmers choose crop types such as maize, rice, cassava, or soybeans.
Farm Input Data	Users enter farm size, estimated input costs, and season duration.
Yield Prediction Model	System calculates predicted yield using crop base yield and investment efficiency.
Loan Request	Farmers request financing based on projected input costs.
Risk Score	Loan risk calculated using loan-to-revenue ratio.
Loan Decision Output	System displays approval, conditional approval, or rejection.

5. MVP Expected Outputs

The system generates three main outputs after a simulation: predicted crop yield, financial risk category, and loan approval decision. Results update instantly when farmers adjust inputs such as crop type or farm size.

6. Growth Stage Product Vision

After validating the MVP, AGRISURE evolves into a full agricultural intelligence platform combining predictive analytics, financing infrastructure, and farmer decision support. The product begins leveraging accumulated agricultural data to improve prediction accuracy and financial risk scoring.

Capability	Description
Weather Simulation	Farmers test yield scenarios under drought, normal rainfall, or heavy rainfall.
Profit Projection	Revenue, cost, and profit estimates calculated using predicted yields.
AI Yield Prediction	Machine learning models trained on historical farm data improve prediction accura
Farmer Credit Scoring	AGRISURE generates a creditworthiness profile used by financial institutions.
Crop Advisory	System recommends better crops or input strategies based on local conditions.
Insurance Integration	Crop insurance providers price policies using platform risk predictions.
Agricultural Data Marketplace	Aggregated anonymized farm data used for national food supply analysis.

7. Success Metrics

Metric	Purpose
Simulation Completion Rate	Measures usability of the prediction tool.
Loan Application Rate	Percentage of farmers requesting financing after simulation.
Prediction Accuracy	Difference between predicted and real yields.
Loan Default Reduction	Measures lender trust and model accuracy.
Farmer Retention	How often farmers return to run new simulations.